

Assignment 4: Novel CNN Architectures for Lightweight Deep Learning

Title:

Implementation of Efficient CNN Models: MobileNet, DenseNet, and EfficientNet

Objective:

Implement modern CNN architectures and evaluate their suitability for resource-constrained systems.

Tasks:

1. Select dataset (MNIST, Fashion-MNIST, CIFAR-10).
2. Implement:
 - MobileNetV2
 - DenseNet121
 - EfficientNetB0
3. Train each model using same settings.
4. Compare:
 - Accuracy
 - Parameters
 - Inference speed
 - Memory usage
5. Test one model on CPU-only environment.
6. Suggest best architecture for:
 - Mobile phone
 - Cloud server
 - Real-time system

Expected Deliverables:

1. Source code
2. Comparison table
3. Performance graphs
4. Final recommendation report

Bonus Task (Optional)

Deploy best model using:

- Flask Web App
- Streamlit
- Gradio Interface